Principles of Economics

Final term :

Topic 4

Money and Monetary Policy

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THE MEANING OF MONEY

- *Money* is the set of assets in an economy that people regularly use to buy goods and services from other people.

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The Functions of Money

- Money has three functions in the economy:
  - Medium of exchange
  - Unit of account
  - Store of value

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The Functions of Money

- Medium of Exchange
  - A *medium of exchange* is an item that buyers give to sellers when they want to purchase goods and services.
  - A medium of exchange is anything that is readily acceptable as payment.

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The Functions of Money

- Unit of Account
  - A *unit of account* is the yardstick people use to post prices.
- Store of Value
  - A *store of value* is an item that people can use to transfer purchasing power from the present to the future.

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Concept of money

- Liquidity
  - *Liquidity* is the ease with which an asset can be converted into the economy's medium of exchange.

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The Kinds of Money

- *Commodity money* takes the form of a commodity with intrinsic value.
  - Examples: Gold, silver

- *Fiat money* is used as money because of government decree.
- It does not have intrinsic value.
- Examples: Coins, currency, check deposits.

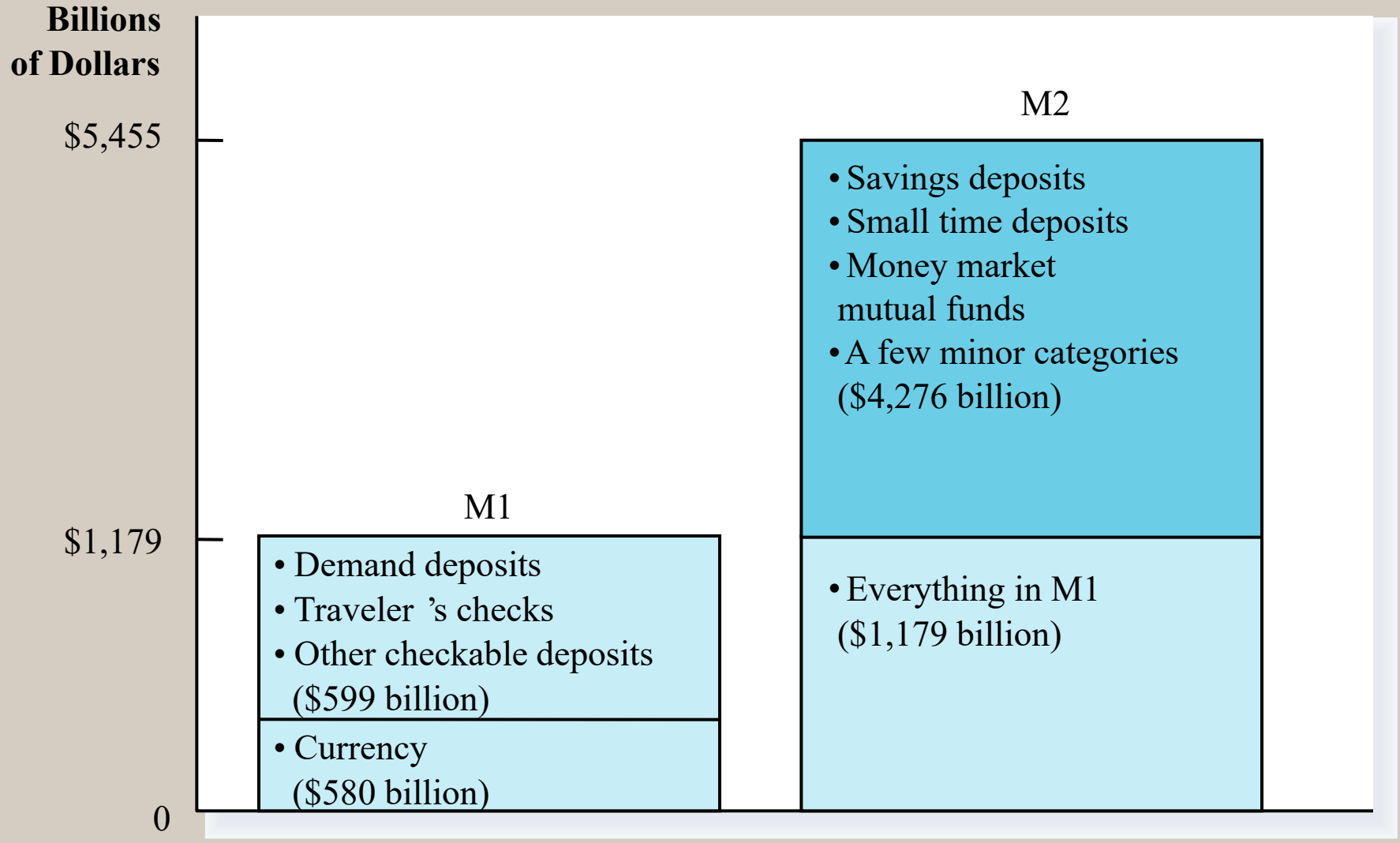
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Money in the Economy

- *Currency* is the paper bills and coins in the hands of the public.
- *Demand deposits* are balances in bank accounts that depositors can access on demand by writing a check.

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Figure 1 Money in the Economy



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BANKS AND THE MONEY SUPPLY

- *Reserves* are deposits that banks have received but have not loaned out.
- In a *fractional-reserve banking* system, banks hold a fraction of the money deposited as reserves and lend out the rest.

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BANKS AND THE MONEY SUPPLY

- Reserve Ratio
  - The *reserve ratio* is the fraction of deposits that banks hold as reserves.

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Money Creation with Fractional-Reserve Banking

- When a bank makes a loan from its reserves, the money supply increases.
- The money supply is affected by the amount deposited in banks and the amount that banks loan.
  - Deposits into a bank are recorded as both assets and liabilities.
  - The fraction of total deposits that a bank has to keep as reserves is called the reserve ratio.
  - Loans become an asset to the bank.

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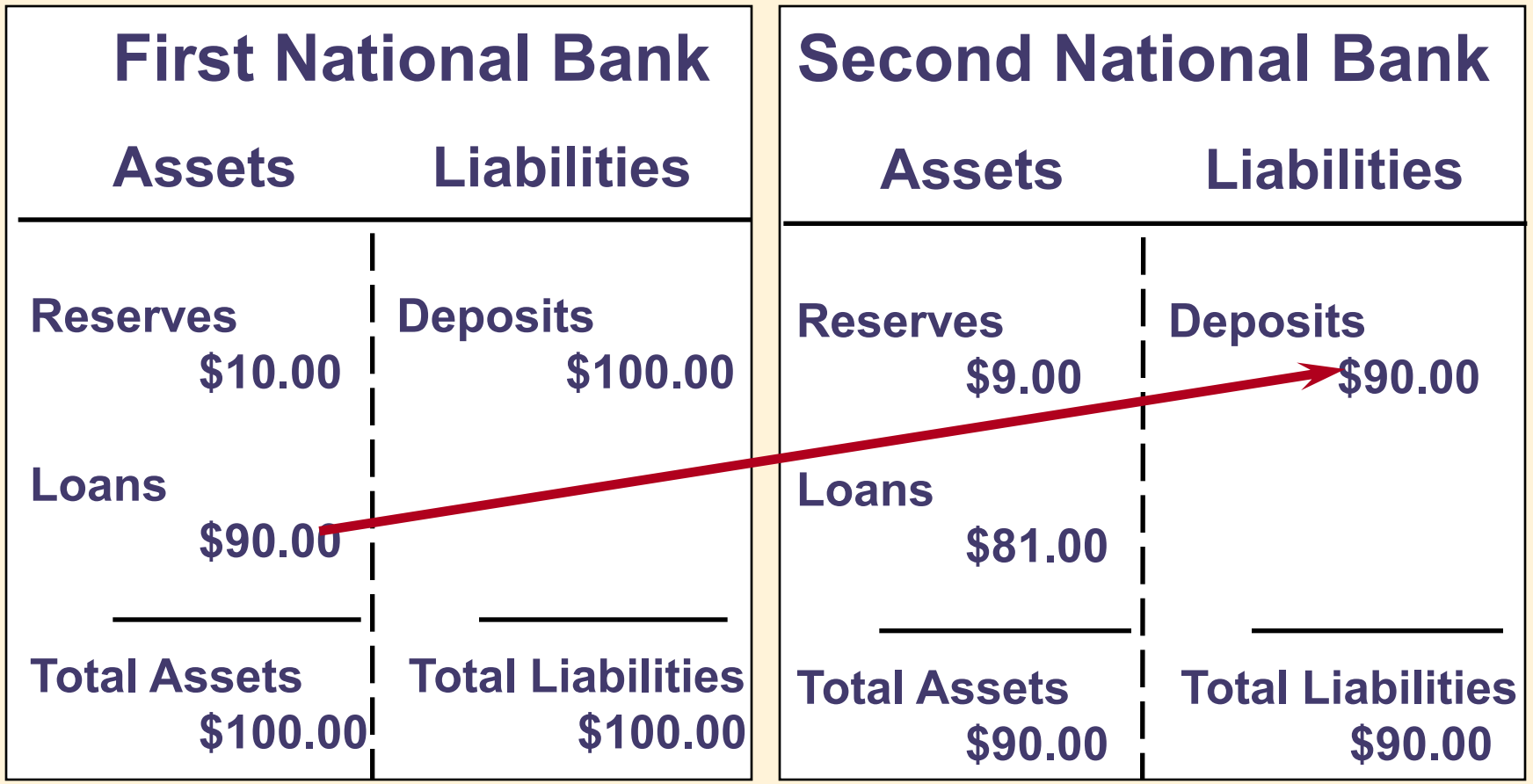
Money Creation with Fractional-Reserve Banking

- This T-Account shows a bank that…
  - accepts deposits,
  - keeps a portion as reserves,
  - and lends out the rest.
  - It assumes a reserve ratio of 10%.

First National Bank

Assets		Liabilities	
Reserves	$10.00	Deposits	$100.00
Loans	$90.00		
Total Assets	$100.00	Total Liabilities	$100.00

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Money Supply = $190.00!

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Money Creation with Fractional-Reserve Banking

- When one bank loans money, that money is generally deposited into another bank.
- This creates more deposits and more reserves to be lent out.
- When a bank makes a loan from its reserves, the money supply increases.

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The Money Multiplier

- How much money is eventually created in this economy?

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The Money Multiplier

- The *money multiplier* is the amount of money the banking system generates with each dollar of reserves.

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The Money Multiplier

- The money multiplier is the reciprocal of the reserve ratio:

$$M = 1/R$$

- With a reserve requirement, $R = 20\%$ or $1/5$,
- The multiplier is 5.

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The Central Bank's Tools of Monetary Control

- The Central Bank has three tools in its monetary toolbox:
  - Open-market operations
  - Changing the reserve requirement
  - Changing the discount rate

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The Central Bank's Tools of Monetary Control

- Open-Market Operations
  - The Central Bank conducts *open-market operations* when it buys government bonds from or sells government bonds to the public:
    - When the Central Bank buys government bonds, the money supply increases.
    - The money supply decreases when the Central Bank sells government bonds.

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The Central Bank's Tools of Monetary Control

- Reserve Requirements
  - The Central Bank also influences the money supply with *reserve requirements*.
  - Reserve requirements are regulations on the minimum amount of reserves that banks must hold against deposits.

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The Central Bank's Tools of Monetary Control

- Changing the Reserve Requirement
  - The *reserve requirement* is the amount (%) of a bank's total reserves that may not be loaned out.
    - Increasing the reserve requirement decreases the money supply.
    - Decreasing the reserve requirement increases the money supply.

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The Central Bank's Tools of Monetary Control

- Changing the Discount Rate
  - The *discount rate* is the interest rate the Central Bank charges banks for loans.
    - Increasing the discount rate decreases the money supply.
    - Decreasing the discount rate increases the money supply.

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Summary

- The term money refers to assets that people regularly use to buy goods and services.
- Money serves three functions in an economy: as a medium of exchange, a unit of account, and a store of value.
- Commodity money is money that has intrinsic value.
- Fiat money is money without intrinsic value.

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Summary

- When banks loan out their deposits, they increase the quantity of money in the economy.
- Because the Central Bank cannot control the amount bankers choose to lend or the amount households choose to deposit in banks, the Central Bank control of the money supply is imperfect.

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